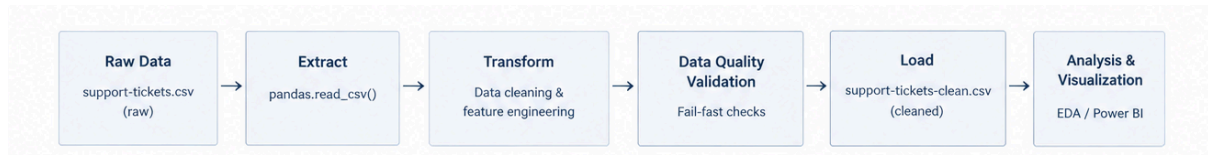


CUSTOMER SUPPORT TICKET ANALYTICS

ETL PROCESS



Python-Based ETL Pipeline

Python was used to build a lightweight ETL pipeline to prepare the raw customer support ticket data before exploratory data analysis and Power BI dashboard development. The pipeline applies the data quality rules defined in `Support_Tickets_Data_Quality_Analysis.ipynb`.

Extract

The raw `support-tickets.csv` file is loaded into a Pandas DataFrame using `pandas.read_csv()`. During ingestion, date fields such as `opened_at` and `resolved_at` are converted into datetime format to support subsequent validation and time-based analysis.

Transform

The raw dataset is transformed through several data preparation steps. These include:

- Standardizing column names and text fields.
- Normalizing categorical values.
- Converting columns to their appropriate data types.
- Deriving analytical fields such as `resolution_min`, `is_open`, `has_csat`, and `opened_month`.
- Deriving `frt_sla_flag` and `resolution_sla_flag` using the IQR-based rule established during the data quality analysis.

These transformations prepare the dataset for consistent analysis while retaining the original ticket-level records.

Data Quality Validation

Before loading the transformed dataset, fail-fast checks are applied to verify that essential data quality requirements are satisfied. The validation focuses on:

- **Required columns:** ensuring all required fields are present.
- **Ticket ID integrity:** checking that `ticket_id` contains no null values and that each ticket ID is unique.
- **Categorical values:** checking and documenting the categorical values used in the dataset.
- **Date consistency:** validating that date and timestamp relationships follow the expected ordering.

The pipeline distinguishes between expected missing values and actual data quality issues. For example, a missing `resolved_at` represents an unresolved ticket and is therefore used to

derive is_open, while a missing csat indicates that a customer satisfaction survey was not completed and is represented through has_csat.

Load

After transformation and successful validation, the prepared dataset is exported as support-tickets-clean.csv. This cleaned dataset is then used as the input for subsequent exploratory data analysis and Power BI dashboard development.

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